25900744 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Multi-Disciplinary Courses (MDC) - MG1MDCBCH100 - SPORTS BIOCHEMISTRY: THE SCIENCE OF EXERCISE AND HUMAN PERFORMANCE

(2024 ADMISSION ONWARDS)

Duration: 1 Hour **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer **All** Questions

Each Question carries **1** mark

1. Which of the following is the first biochemical event in tissue repair follows a sports injury? [K] / [CO5]

   a). Clot formation b). Inflammatory response

   c). Fibroblast activation d). Muscle regeneration

2. Which of the following functions do nutrients in food serve in the body? [U] / [CO2]

   a). Build and repair tissues b). Provide energy

   c). Regulate body process d). All the above

3. In high-intensity intermittent sports, what energy system is predominantly used? [U] / [CO3]

   a). Oxidative system b). ATP-PC system

   c). Anaerobic glycolysis d). Fatty acid oxidation

4. What is the primary goal of an individualized training program for an athlete? [U] / [CO4]

a). follow a one-size-fits-all approach for all athletes

b). focus solely on improving cardiovascular fitness

c). design a training plan that addresses the athlete's specific needs, goals, and physical condition

d). increase muscle mass without considering injury prevention

5. Which of the following is a key enzyme released during muscle damage? [U] / [CO5]

a). Troponin

b). Creatine kinase

c). C-reactive protein

d). Myoglobin

[5x1 = 5]

Part B

Fill in the Blanks

Answer **All** Questions

Each Question carries **2** marks

6. Motion capture systems help to visualisethe athlete's movement and quantify [U] / [CO4]

7. In sports biochemistry,…….refers to the study of how genes and physical activity interact to influence athletic performance. [U] / [CO1]

8. The energy system active for exercise of more than two minutes duration is_____ [K] / [CO3]

9. _____ involves visualizing successful outcomes to improve motivation and reduce anxiety before events. [U] / [CO5]

10. The _____ phase of sleep is most important for physical recovery. [U] / [CO5]

[5x2 = 10]

Part C

Short Answer Questions

Answer any **5** Questions

Each Question carries **3** marks

11. Recall hydration and hydration strategies during exercise. [K] / [CO2]

12. Illustrate the main source for replenishing glycogen after exercise [U] / [CO5]

13. Evaluate the roles of skeletal muscles in the human body? [E] / [CO3]

14. Why is recovery essential in preventing sports injuries? [U] / [CO5]

15. Design a weekly recovery plan for an athlete based on HRV and sleep tracking data. [Ap] / [CO4]

16. Explain the biochemical effects of thermotherapy [E] / [CO5]

17. What is the difference between a sprain and a strain? [E] / [CO5]

[5x3 = 15]

Part D

Short Essay Type Questions

Answer any **1** Question

Each question carries **5** marks

18. Evaluate the classification of ergogenic aids. [U] / [CO4]

19. Illustrate the influence of hormones in sports and exercise [An] / [CO3]

[1x5 = 5]